# STOCK MARKET VOLATILITY & FINANCIAL ASSET FORECASTING

## Introduction

This project uses past stock market data and machine learning to predict future stock prices, estimate market risk, and predict whether stock prices will go up or down.

Stock prices can change because of many factors, such as company performance, trading activity, interest rates, inflation, investor opinions, and unexpected events. Because of these factors, predicting stock prices accurately can be difficult.

**GOAL:**

This project is not just about creating charts or calculating average prices. It uses supervised machine learning, where past market data is used to train a model. The model learns from this data and tries to predict future results. Finally, its predictions are tested using newer data that was not used during training to check how well the model performs.

## Understanding the Business Objective

This project has three main goals:

**1. Price Prediction (Regression):** Predict the future stock price, such as tomorrow's closing price or the closing price after five trading days.

**2. Price Direction Prediction (Classification):** Predict whether the stock price will go up or down in the future.

**3. Volatility Prediction:** Predict how much the stock price may change over a period of time. This helps us understand the possible risk and uncertainty in the market.

The project also aims to compare the model's predictions with simple prediction methods and identify which factors are most important for making accurate predictions.

In simple words: The main goal is to predict future stock prices, identify whether prices will rise or fall, understand market risk, and check how well the model performs compared to simpler methods.

**What do the important words mean?**

| Term | Simple meaning |
|---|---|
| **Financial asset** | A stock, index, or other financial instrument |

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| **Volatility** | How much and how quickly prices fluctuate |
| **Forecasting** | Predicting a future value |
| **Dataset** | A collection of information, usually stored in a CSV or Excel file |
| **Regression** | Predicting a number, such as a future stock price |
| **Classification** | Predicting a category, such as price going up or down |
| **Feature** | Information given to the model, such as past price and volume |
| **Target** | The value the model is supposed to predict |

# Our Interpretation and Thought Process

From my understanding, this project aims to use historical stock market data to build a machine learning model that can help predict future stock prices or price movements.

we understood that the model will learn from past stock prices, trading volume, and other relevant indicators. After training, its predictions will be compared with actual stock market data to evaluate its performance.

we also understood that the project is not only about making predictions but also about identifying the important factors that influence those predictions and understanding the risks associated with market volatility.

# Assumption to solve the problem

We plan to begin by finding a suitable historical stock market dataset from a reliable source. We will examine the dataset to understand its columns, time period, and data quality.

Next, we will prepare the data and identify useful features such as previous closing prices, trading volume, and technical indicators. we will then select a suitable prediction target and train a machine learning model.

Finally, we will compare the model's predictions with actual values and a simple baseline method to understand how well it performs.

# Dataset research

We searched relevant data set in KAGGLE

We found a data set

The dataset contains a total of 25,161 rows, each row representing the stock market data for a specific company on a given date. The information collected through web scraping from [www.nasdaq.com](http://www.nasdaq.com)

includes the stock prices and trading volumes for the companies listed, such as Apple, Starbucks, Microsoft, Cisco Systems, Qualcomm, Meta, Amazon.com, Tesla, Advanced Micro Devices, and Netflix.

**Dataset Name:** Stock Market: Historical Data of Top 10 Companies

**Dataset Source:** Kaggle

**Number of Records:** 25,161 rows

**Data Description:** The dataset contains historical stock market information for multiple companies, including Apple, Microsoft, Amazon, Tesla, and other companies.

**Important Features:** The dataset is expected to contain stock-related information such as dates, opening prices, highest prices, lowest prices, closing prices, and trading volume. These columns must be verified by inspecting the downloaded dataset.

**Reason for Selection:** This dataset may be suitable for the project because historical stock prices and trading volumes can be used to create features for predicting future prices or price direction.

# Expected Outcome

The expected outcome of this project is to develop and evaluate a machine learning model for financial forecasting. The project aims to understand whether historical market data can help predict future stock prices or price direction.

It will also help identify useful market indicators, compare machine learning predictions with simple forecasting methods, and understand the limitations of predicting stock market movements.

# Challenges and Limitations

Stock market prediction is challenging because prices are affected by many changing factors and unexpected events. Historical patterns may not continue in the future, and the available data may contain missing values or noise.

Another challenge is ensuring that the model uses only information available at the time of prediction. Therefore, model performance must be evaluated carefully using data from a later time period.

# Questions we got in research

**Research Questions**

1. Can historical stock prices and trading volumes help predict future stock prices?
2. Which features are most useful for predicting future price movements?
3. Can machine learning models perform better than simple forecasting methods?
4. How accurately can the model classify whether a stock price will rise or fall?

5. What challenges affect the reliability of stock market predictions?

## Source we used

We used CHAT GPT, KAGGLE, META AI, GOOGLE

## Conclusion

From this preliminary analysis, we understand that the project aims to apply machine learning techniques to historical stock market data to predict future prices or price movements. The next step is to examine the selected dataset, identify suitable features and prediction targets, and determine which machine learning approach is appropriate. The model's performance will be evaluated against simple forecasting baselines to understand its effectiveness and limitations.